

Course: B.Tech

Branch : Information Technology

Semester : III

Subject Code & Name: BTITC305 Data Structures and Applications

Max Marks: 60

Date:21/02/2025

Duration: 03 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

Q. 1 Objective type questions. (Compulsory Question)**12**

- | | Level
(CO) | Marks |
|--|---------------|-------|
| 1. Pick up the not related characteristic of an Algorithm: | CO1 | 1 |
| a. It is a formal step by step representation of given problem statement
b. It require mathematical function/model to calculate time complexity.
c. 8/0 represent the precise line of code.
d. It will produces the desired output with respect to given input. | | |
| 2. Which of the following asymptotic notation is used to provide lower bound constraint. | CO3 | 1 |
| a. O (Big-Oh)
b. Ω (Big-Omega)
c. Θ (Big-Theta)
d. None of the above | | |
| 3. The result evaluating the postfix expression $10\ 5 + 60\ 6 / * 8 -$ is | CO2 | 1 |
| a. 284
b. 213
c. 71
d. 142 | | |
| 4. What condition should be checked before the deletion operation from a Queue? | CO4 | 1 |
| a. Front value
b. Rear value
c. Underflow
d. Overflow | | |
| 5. What is the output of the program after performing the following functions? | CO5 | 1 |
| <pre> push(10); push(20); pop(); push(15); pop(); </pre> | | |

pop();
peek();

- a. 20 b. 10 c. 15 d. Stack Underflow

6. Consider the following numbers 8, 1, 2, 4, 3, 9, 6, 5 to be inserted into the empty binary search tree. What will be the sequence of in-order traversal of the resultant binary search tree? C04 1

- a. 8 1 3 5 6 9 2 4 b. 1 2 3 4 5 6 8 9 c. 5 6 9 3 4 2 1 8 d. 8 1 2 4 3 9 6 5

7. Consider the following statements. C04 1
S₁: The sequence of procedure calls corresponds to a pre-order traversal of the activation tree.

S₂: The sequence of procedure returns corresponds to a post-order traversal of the activation tree.

Which one of the following options is correct?

- a. S₁ is true and S₂ is true b. S₁ is false and S₂ is true c. S₁ is true and S₂ is false d. S₁ is false and S₂ is false

8. Which statement/s is/are true for the graph? C03 1
a. It may or may not contain cycle. b. It can identify the boundaries of the regions. c. Minimum Spanning tree is the application of it. d. All of the above.

9. Which data structure is used to perform level-order traversal on a binary tree? C03 1
a. Stack b. Linked List c. Queue d. Hash Table

10. Which statement/s is/are true for the Quick Sort? C05 1
a. It's worst case time complexity is O(n*n) b. It divide the array by the calculation of mid. c. It require the iterative method to sort the given input array. d. All of the above.

11.	Choose the incorrect choice/s for Recursive method of Binary Search	CO5	1
	a. It doesn't require the calculation of mid element. b. If element is found then it always present at mid. c. It require the input as a sorted array d. None of the above		
12.	Pickup correct statement	CO1	1
	a. The stack and queue are used in graph traversal as BFS and DFS respectively. b. Stack used in reversing the input string. c. Postfix Expression evaluation carried by stack d. Both b and c		
Q. 2	Solve the following.		12
A)	Define Algorithm and Explain any two asymptotic notation with neat diagram.	CO2	6
B)	Define Recursion and explain it with an example.	CO5	6
Q.3	Solve the following.		12
A)	Explain Circular queue with neat diagram and its need.	CO3	6
B)	Write an algorithm to delete a node from the end of singly linked list (SLL).	CO4	6
Q. 4	Solve Any Two of the following.		12
A)	Define Binary tree. Construct the binary tree for the following data 22, 16, 5, 9, 27, 105, 12, 4, 7. Also calculate post order of the constructed binary tree.	CO2	6
B)	Define Binary Search tree and perform the following operation (Root: 25) i) Delete 30 ii) Delete 20	CO4	6

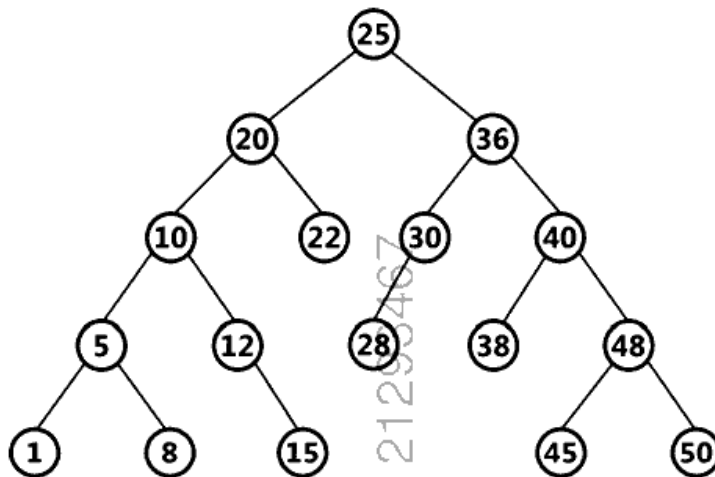


Fig. 1

- C) Explain Array representation of Binary tree. Show the array representation of fig.1

CO3

6

Q.5 Solve Any Two of the following.

12

- A) Define graph and explain it by adjacency matrix representation with an example.
 B) Construct the Minimum spanning tree in stepwise manner by using Prim's algorithm for the following graph. (Source: 0)

CO1

6

CO4

6

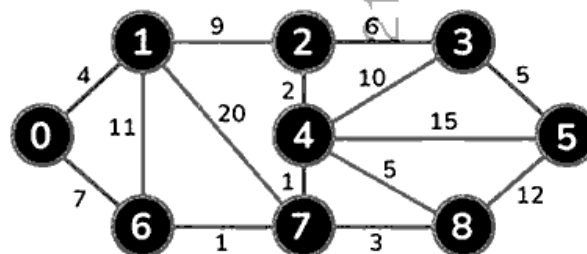


Fig.2

- C) Differentiate between Breadth- First-Search and Depth-First- Search method.

CO5

6

Q. 6 Solve Any Two of the following.

12

- A) Explain Binary Search with an example.
 B) Explain Quick sort and what is role of pivot element in quick sort.
 C) Write a short note on Heap sort.

CO2

6

CO3

6

CO4

6

*** End ***